



TurnZero: The Illusion of Adaptation When Experts Lose

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Problem: losers change, but does it help?

Pokémon VGC best-of-three: both players see each other's full roster (**Open Team Sheets**), then privately pick 4-of-6 to bring and 2-of-4 to lead. $C(6,4) \times C(4,2) = \mathbf{90\text{-way decision}}$ before any battle begins. We study **212k tournament battles (382k directed examples)**.

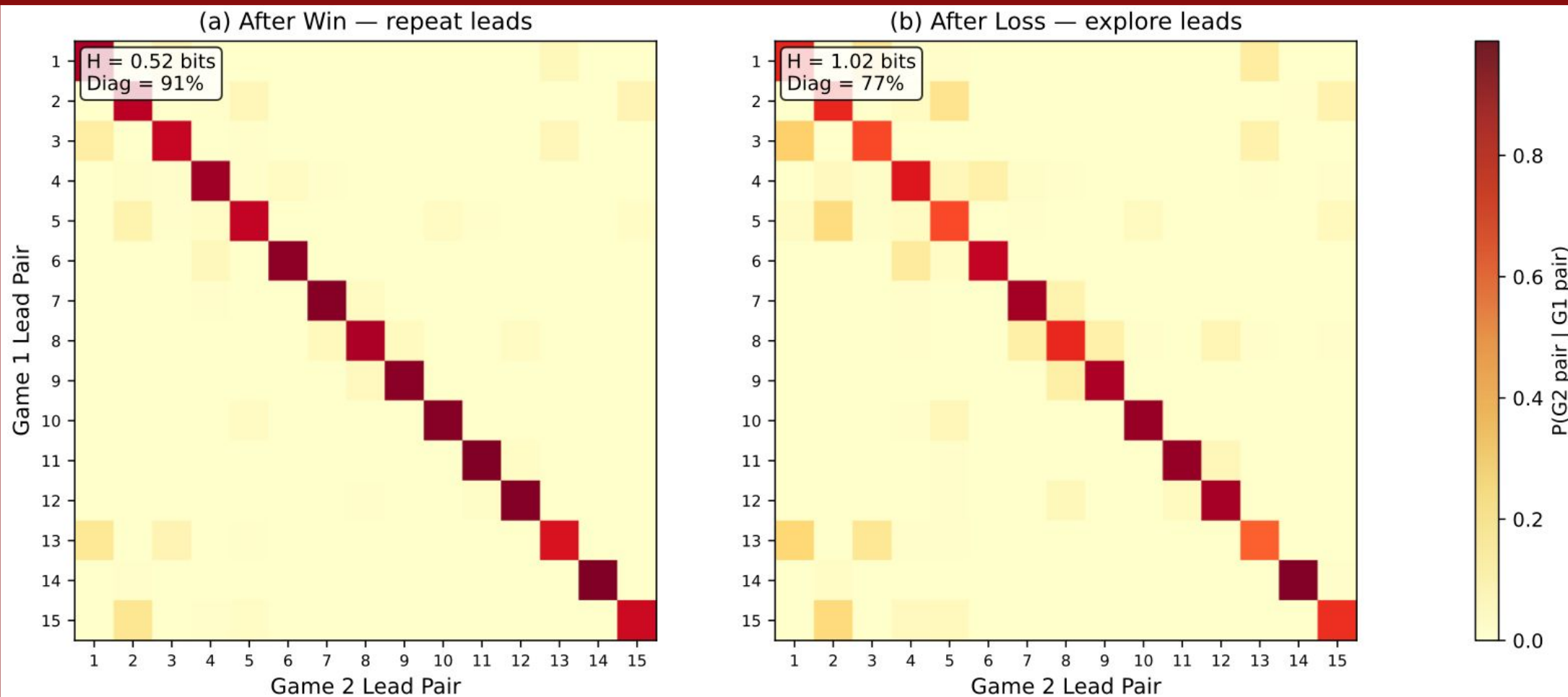
Players who just lost change their leads 72% of the time vs. 46% for winners, which looks like adaptation. But leads are the most visible lever to pull after a loss, not necessarily the cause of it. We use transformer ensembles to measure. A sequential model gains **+6.6pp after wins but <1pp after losses**. A specialist trained only on post-loss data scores below baseline (4.0% vs. 5.4%). **No model can learn to predict post-loss behavior**.

Data & Features

212K tournament battles from Pokémon Showdown, yielding **382K directed examples** and 136K game-to-game transitions.

Each example pairs two Open Team Sheets with a ground-truth 90-way action. Per Pokémon: **8 features (species, item, ability, tera type, 4 moves)**, all learned via embeddings.

Winners stay, losers scatter



Transition matrices for 15 most common lead pairs. After winning: 91% diagonal, $H=0.52$ nats. After losing: 77% diagonal, $H=1.02$ nats

	After win	After loss
Change rate	46%	72%
Diagonal mass	91%	77%
Entropy (nats)	0.52	1.02

Losers spread across 2x more of the action space, but without direction.

Exploration is unstructured

If post-loss changes were counter-play, they should respond to what the opponent did.

- **87%** of post-loss lead switches don't change overlap with the opponent's prior leads ($n = 43,063$), yet win G2 at 58.2%
- The 13% that do: switching toward and away produce **identical G2 win rates** (54.8% vs. 54.9%)

The 27pp gap between change rates holds across all Elo quartiles.

Elo tier	After loss	After win	Gap
Q1 (low, <1119)	78.6%	53.3%	25.3pp
Q2 (1119–1197)	75.0%	46.9%	28.0pp
Q3 (1197–1303)	69.9%	42.6%	27.3pp
Q4 (high, >1303)	64.3%	37.3%	27.1pp

Models learn convention, not strategy

5-member deep ensemble of permutation-invariant transformers (4 layers, $d=128$, $H=4$). Each Pokémon tokenized as the sum of 8 learned embeddings, canonically sorted, mean-pooled. **5.8M params**. ECE = 0.012.

$$\hat{p}(a) = \frac{1}{M} \sum_{m=1}^M p_{\theta_m}(a \mid \mathbf{x}_{\text{self}}, \mathbf{x}_{\text{opp}}), \quad M = 5$$

6.4% top-1 on the 90-way task (vs. 1.1% random). Removing the opponent at inference costs only 1.3pp; removing the player's own team collapses to random. **The model learns what teams usually do, not how they adapt**.

History helps winners, not losers

A sequential variant prepends a context token (game number, prior result, prior lead pair). Only +2,816 params (<0.3% of base).

Game-1 accuracy remains unchanged, so no leakage.

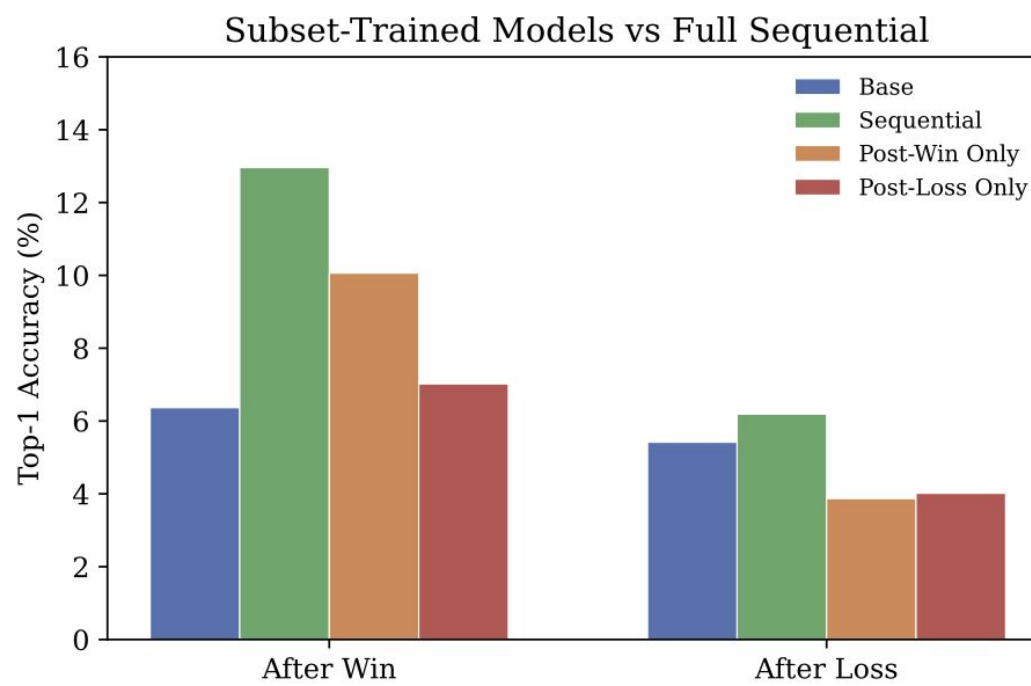
Model	Params	After win	After loss	G2–3
Base ensemble	5.8M	6.4	5.4	5.9
Sequential	+2.8K	13.0	6.2	9.4
+ Opp. context	+4.1K	12.6	6.0	9.2
Post-win spec.	+2.8K	10.1	3.9	—
Post-loss spec.	+2.8K	7.0	4.0	—

+6.6pp after wins. +0.8pp after losses. **After losses, there's nothing to learn in the data**.

Post-loss is irreducible

Three experiments confirm post-loss behavior has no structure:

- **Adding opponent's revealed species** (+4.1K params) produces no post-loss improvement and slightly hurts post-win (12.6% vs. 13.0%)
- A **model trained only on post-loss data** scores 4.0% on its own target data, below the base model's 5.4% on the same subset. The signal actively misleads.
- A **model trained only on post-win data** reaches 10.1% after wins. Post-win behavior has structure missing from post-loss.



Discussion

Winners have confirmed info ("this worked") and repeat it.

Losers can't isolate why they lost. A battle involves dozens of micro-decisions, and leads are rarely the true cause. So 72% change, but each loser scrambles in their own direction. In a 90-way space, it all washes out.

Conclusions & Future Work

- **Win-stay is rational**: +6.6pp, requires only memory
- **Lose-shift is noise**: undirected shifts in a 90-way space produce no structure
- **The base model learns convention, not strategy** (opponent ablation: only 1.3pp)
- **Post-loss signal misleads**: specialist goes below baseline (4.0% vs. 5.4%)

Going forward

- Replicate across other VGC regulations and competitive domains with combinatorial team selection (MOBA drafts, fighting game counterpicks)
- Randomized lead assignment in simulation to test causally whether post-loss adaptation helps

References

- [1] Nowak & Sigmund, "Win-stay, lose-shift," *Nature*, 1993.
- [2] Angliss et al., "VGC-Bench," AAMAS, 2026.
- [3] Lakshminarayanan et al., "Deep Ensembles," *NeurIPS*, 2017.